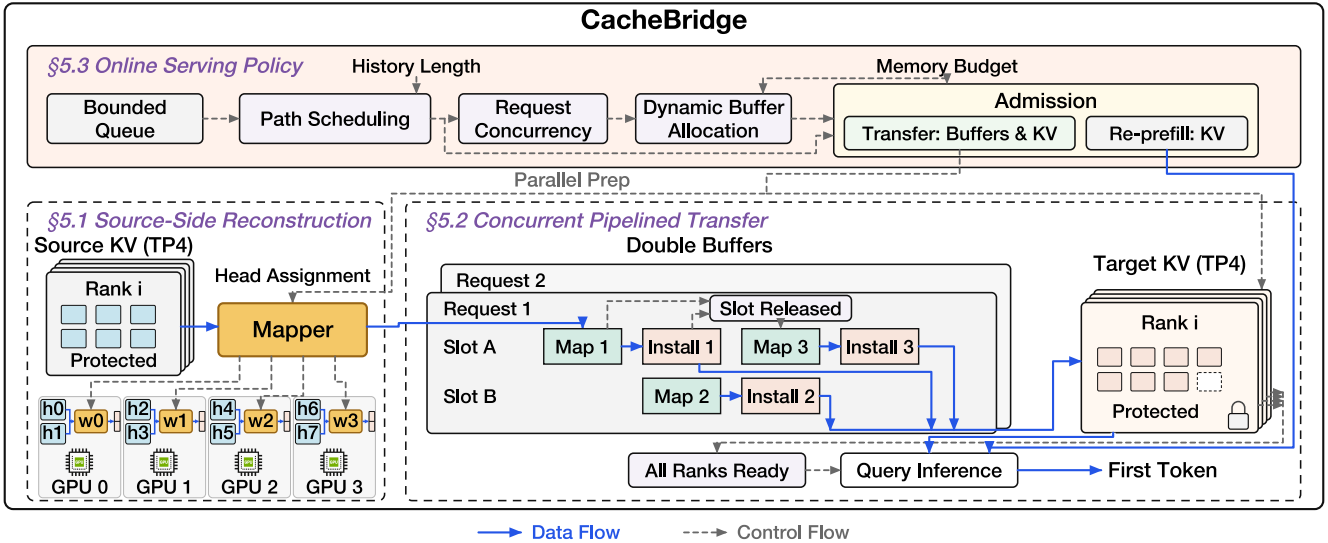

CacheBridge
§5.3 Online Serving Policy
History Length
Memory Budget
Bounded Queue
Path Scheduling
Request Concurrency
Dynamic Buffer Allocation
Admission
Transfer: Buffers & KV
Re-prefill: KV
Parallel Prep
§5.1 Source-Side Reconstruction
Source KV (TP4)
Rank i
Protected
Head Assignment
Mapper
h0
h1
w0
h2
h3
w1
h4
h5
w2
h6
h7
w3
GPU 0
GPU 1
GPU 2
GPU 3
§5.2 Concurrent Pipelined Transfer
Double Buffers
Request 2
Request 1
Slot Released
Slot A
Map 1
Install 1
Map 3
Install 3
Slot B
Map 2
Install 2
Target KV (TP4)
Rank i
Protected
All Ranks Ready
Query Inference
First Token
Data Flow
Control Flow